

HALD VEST, Denmark

WMO: 060490

Lat: 56.561N

Lon: 10.093E

Elev: 292

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.1	(c) 21.2	(d) 13.1	(e) 10.8	(f) 20.8	(g) 16.2	(h) 12.6	(i) 23.6	(j) 45.3	(k) 40.9	(l) 36.4	(m) 40.9	(n) 8.3	(o) 350	(p) 0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	78.1	64.9	74.9	63.6	71.8	62.1	66.7	74.3	65.0	72.0	63.4	69.7	10.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	89.9	69.2	62.3	85.0	67.8	60.8	80.3	66.5	31.4	74.5	30.1	71.9	28.9	69.9	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 31.7	(b) 26.1	(c) 22.7	(e) 15.7	(f) 83.4	(g) 3.7	(h) 2.9	(i) 13.0	(j) 85.5	(k) 10.8	(l) 87.2	(m) 8.8	(n) 88.9	(o) 6.1	(p) 91.0		
(a) 31.7	(b) 26.1	(c) 22.7	(e) 14.4	(f) 69.5	(g) 4.4	(h) 1.5	(i) 11.2	(j) 70.7	(k) 8.7	(l) 71.6	(m) 6.2	(n) 72.4	(o) 3.0	(p) 73.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	33.1	33.4	37.6	44.9	51.9	56.9	61.8	61.6	56.0	48.4	40.9	35.9
	DBStd	11.70	6.50	6.26	6.07	5.73	5.82	4.86	5.07	4.52	4.55	4.99	5.38	6.65
	HDD50	2397	525	465	384	171	46	3	0	0	4	89	274	436
	HDD65	6641	990	885	848	602	406	247	130	126	271	513	722	901
	CDD50	1291	0	0	1	19	106	211	365	361	185	41	2	0
	CDD65	60	0	0	0	0	1	5	30	22	2	0	0	0
	CDH74	353	0	0	0	1	2	36	202	107	5	0	0	0
	CDH80	34	0	0	0	0	0	2	22	10	0	0	0	0
Wind	WSAvg	11.4	13.3	13.1	12.7	11.4	10.5	10.2	9.0	9.7	10.6	11.7	12.2	12.4
Precipitation	PrecAvg	27.6	2.3	1.7	1.8	1.6	2.0	2.4	2.5	3.0	2.6	2.9	2.5	2.4
	PrecMax	35.1	4.2	4.6	3.8	3.7	3.8	5.3	4.6	6.6	7.4	5.5	5.0	4.8
	PrecMin	19.5	0.2	0.4	0.2	0.2	0.3	0.1	0.6	0.9	0.6	0.6	0.9	0.8
	PrecStd	3.9	1.1	0.9	0.9	1.0	0.8	1.1	1.1	1.4	1.5	1.2	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	49.5	55.7	66.1	73.1	78.5	83.1	81.4	74.2	62.6	55.0	51.0
		MCWB	47.0	47.1	48.0	55.2	61.7	65.0	66.2	65.7	62.2	58.0	52.9	49.2
	2%	DB	45.9	46.1	51.3	61.3	69.3	73.7	79.0	76.9	69.9	59.4	52.5	48.6
		MCWB	44.5	44.0	46.2	52.0	59.7	63.0	65.1	64.4	61.1	56.1	50.8	47.2
	5%	DB	44.0	44.0	48.1	57.9	65.5	70.2	76.0	73.5	66.3	57.3	50.4	46.5
		MCWB	42.7	42.4	43.8	49.9	57.0	61.3	64.5	63.2	59.1	54.7	48.9	45.4
	10%	DB	42.3	42.0	45.5	54.3	62.3	66.5	72.4	70.1	63.4	55.4	48.6	44.9
		MCWB	41.0	40.6	42.5	48.3	55.5	59.0	62.8	61.2	57.8	53.4	47.3	43.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.5	47.4	49.9	57.1	64.1	67.2	69.1	68.2	64.3	59.0	53.5	49.5
		MCDB	48.5	48.7	53.9	63.3	71.1	74.6	77.8	75.5	70.4	61.3	54.7	50.4
	2%	WB	44.8	44.6	46.8	53.2	60.7	64.4	67.3	66.1	61.9	56.9	51.0	47.4
		MCDB	45.7	45.6	50.3	58.9	66.5	72.0	75.5	73.4	67.9	58.5	52.3	48.2
	5%	WB	42.9	42.5	44.7	50.8	58.4	62.1	65.5	64.4	60.3	55.0	49.2	45.7
		MCDB	43.7	43.7	47.1	56.0	64.2	68.5	72.9	71.0	65.3	56.7	50.2	46.5
	10%	WB	41.1	40.7	43.0	48.8	56.2	59.8	63.7	62.8	58.7	53.6	47.4	43.9
		MCDB	42.0	41.9	45.1	53.6	61.3	65.1	70.4	68.0	62.2	55.3	48.5	44.8
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.8	9.8	13.0	14.6	15.0	15.6	14.6	12.2	8.7	7.1	6.5
		MCDBR	7.8	8.5	14.1	17.7	18.6	19.5	20.1	18.5	15.6	10.1	8.3	7.5
		MCWBR	7.8	7.2	9.6	10.7	11.6	11.5	9.4	8.3	8.4	7.4	7.7	7.4
	5% WB	MCDBR	7.7	8.1	13.0	16.2	16.8	18.0	18.2	16.1	14.1	9.5	8.0	7.4
		MCWBR	7.8	7.3	9.4	10.5	11.4	11.6	9.7	8.5	8.4	7.6	7.5	7.2
Clear-Sky Solar Irradiance	taub	0.307	0.330	0.349	0.385	0.378	0.370	0.389	0.390	0.364	0.345	0.323	0.292	
	taud	2.241	2.296	2.321	2.277	2.343	2.388	2.356	2.365	2.424	2.409	2.303	2.218	
	Ebn at Noon	181	229	256	263	273	277	268	259	250	222	174	153	
	Edh at Noon	17	24	30	36	36	35	36	33	27	22	16	13	
All-Sky Solar Radiation	RadAvg	133	315	761	1277	1685	1825	1680	1323	915	452	171	95	
	RadStd	26	52	70	150	151	144	178	103	94	56	28	12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	-212	-295	N/A	N/A

Nomenclature: See separate page